PARENT ONLY — SOLVED ANSWER KEY

Vyom · Parent answer key

22 Feb – 28 Feb 2027 · Monday–Sunday · NDPS Class 9

Vyom ko test se pehle ye PDF mat dena. Student pack mein sirf questions aur blank answer space hai. Is alag parent PDF mein har original question ke saath uska matching solved answer, working aur marking points diye hain; Test ID se student paper match karein.

Checking: same evening. C = concept, R = recall, K = careless, T = time.

90%+: short revision. 75–89%: weak-topic practice + retest. Below 75%: topic dobara samjhao; mistakes notebook mein likho.

Answer keys existing source question banks se li gayi hain; school pattern abhi unconfirmed hai. Alternate valid answers ko marking guidance ke mutabik accept karein.

Week score register

Date / test ID	Test	Marks	Galti / retest
22 Feb T2027-02-22	Maths Mock (2-hr section practice)	___ / 80	Retest: 25 Feb
23 Feb T2027-02-23	Science Mock (2-hr section practice)	___ / 80	Retest: 26 Feb
24 Feb T2027-02-24	SST Mock (2-hr section practice)	___ / 80	Retest: 27 Feb
25 Feb T2027-02-25	Hindi Mock (2-hr section practice)	___ / 80	Retest: 28 Feb
26 Feb T2027-02-26	English Mock (2-hr section practice)	___ / 80	Retest: 1 Mar
27 Feb T2027-02-27	Maths Mock (2-hr section practice)	___ / 80	Retest: 2 Mar
28 Feb T2027-02-28	Maths Full paper (3 hr)	___ / 80	Retest: 3 Mar

Maths · Mock Paper 2 · Answer key

Monday, 22 February 2027 · 80 marks

Test ID: T2027-02-22

Class 9 · poora syllabus (Ch 1–14)

Q1 **[1]**

Question: The point (–4, –4) lies in quadrant:

(a) I (b) II

(c) III (d) IV

Expected answer / marking points: (c) III

Q2 **[1]**

Question: The distance between (–3, 2) and (5, 2) is:

(a) 2 (b) 8

(c) 7 (d) 3

Expected answer / marking points: (b) 8

Q3 **[1]**

Question: The linear polynomial with slope 4 and y-intercept –1 is:

(a) –x + 4 (b) 4x – 1

(c) x – 4 (d) 4x + 1

Expected answer / marking points: (b) 4x – 1

Q4 **[1]**

Question: The value of x for which 5x – 10 = 0 is:

(a) 5 (b) –2

(c) 2 (d) 10

Expected answer / marking points: (c) 2

Q5 **[1]**

Question: Which of the following is rational?

(a) $\sqrt{2}$ (b) $\sqrt{9}$

(c) π (d) $\sqrt{5}$

Expected answer / marking points: (b) $\sqrt{9}$

Q6 **[1]**

Question: $(2\sqrt{3})^2$ =

(a) 6 (b) 12

(c) $4\sqrt{3}$ (d) 36

Expected answer / marking points: (b) 12

Q7 **[1]**

Question: The coefficient of x in (x + 4)(x – 7) is:

(a) –3 (b) 3

(c) –28 (d) 11

Expected answer / marking points: (a) –3

Q8 [1]

Question: $(a + b)^3 - (a - b)^3 =$

(a) $6a^2b + 2b^3$ (b) $2a^3$

(c) $6ab^2$ (d) 0

Expected answer / marking points: (a) $6a^2b + 2b^3$

Q9 [1]

Question: How many circles can pass through three non-collinear points?

(a) None (b) Exactly one

(c) Two (d) Infinitely many

Expected answer / marking points: (b) Exactly one

Q10 [1]

Question: The length of a semicircular arc of radius 14 cm is:

(a) 22 cm (b) 44 cm

(c) 88 cm (d) 28 cm

Expected answer / marking points: (b) 44 cm

Q11 [1]

Question: A die is thrown once. The probability of NOT getting a 6 is:

(a) 1/6 (b) 5/6

(c) 1 (d) 0

Expected answer / marking points: (b) 5/6

Q12 [1]

Question: If $t_n = 3n + 2$, the common difference of the AP is:

(a) 2 (b) 3

(c) 5 (d) 6

Expected answer / marking points: (b) 3

Q13 [1]

Question: The 5th term of the GP 2, –6, 18, ... is:

(a) –162 (b) 162

(c) 54 (d) –54

Expected answer / marking points: (b) 162

Q14 [1]

Question: "If a quadrilateral is a square, then it is a rectangle." Which is correct?

(a) Proposition true, converse true

(b) Proposition true, converse false

(c) Proposition false, converse true

(d) Both false

Expected answer / marking points: (b) Proposition true, converse false

Q15 [1]

Question: 2 L of 40% juice is mixed with 2 L of 10% juice. The concentration of the mixture is:

(a) 50% (b) 25%

(c) 30% (d) 20%

Expected answer / marking points: (b) 25%

Q16 [1] **Question:** To test whether 49 is prime, an algorithm checks divisors 2, 3, ..., 7. How many checks is that?

(a) 5

(b) 6

(c) 7

(d) 49

Expected answer / marking points: (b) 6

Q17 [1] **Question:** The diagonals of a rectangle are:

(a) perpendicular (b) equal

(c) unequal (d) parallel

Expected answer / marking points: (b) equal

Q18 [1] **Question:** The curved surface area of a cylinder of radius 7 cm and height 10 cm is:

(a) 220 cm^2 (b) 440 cm^2

(c) 1540 cm^2 (d) 308 cm^2

Expected answer / marking points: (b) 440 cm^2

Q19 [1] **Question: A:** The angle in a semicircle is a right angle.
R: The angle subtended by an arc at the centre is double the angle subtended at any point on the remaining part of the circle.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q20 [1] **Question: A:** 0.101001000100001... is irrational.
R: Its decimal expansion is non-terminating and non-repeating.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q21 [2] **Question:** Three vertices of a square are (1, 1), (4, 1) and (4, 4). Find the fourth vertex and the area of the square.

Expected answer / marking points: (1, 4) [1]; side 3, area 9 sq units [1].

Q22 [2] **Question:** Evaluate 47 × 53 using a suitable identity.

Expected answer / marking points: (50 – 3)(50 + 3) = 2500 – 9 [1] = 2491 [1].

Q23 [2] **Question:** Find the area of a triangle with sides 5 cm, 12 cm and 13 cm using Heron's formula.

Expected answer / marking points: s = 15 [0.5]; $\sqrt{(15 \times 10 \times 3 \times 2)} = \sqrt{900}$ [1] = 30 cm^2 [0.5].

Q24 **[2]** **Question:** A family has two children. Write the sample space (B = boy, G = girl) and find P(at least one girl).

Expected answer / marking points: S = {BB, BG, GB, GG} [1]; 3/4 [1].

Q25 **[2]** **Question:** Is 301 a term of the AP 5, 11, 17, ...? Justify.

Expected answer / marking points: $5 + (n - 1)6 = 301 \Rightarrow n - 1 = 296/6 = 49\frac{1}{3}$ [1]; not a whole number, so 301 is not a term [1].

Q26 **[3]** **Question:** Prove that the sum of a rational number and an irrational number is irrational.

Expected answer / marking points: Let r rational, s irrational, and suppose $r + s = q$ is rational [1]; then $s = q - r$ is rational (difference of rationals) [1]; contradiction $\Rightarrow$ $r + s$ is irrational [1].

Q27 **[3]** **Question:** A chord AB of a circle of radius 10 cm subtends a right angle at the centre. Find the length of AB and its distance from the centre.

Expected answer / marking points: $AB = \sqrt{(10^2 + 10^2)} = 10\sqrt{2} \approx 14.14$ cm [1.5]; distance $= \sqrt{(100 - 50)} = 5\sqrt{2} \approx 7.07$ cm [1.5].

Q28 **[3]** **Question:** A pendulum 21 cm long swings through an angle of 60°. Find the length of the arc its tip describes and the area it sweeps.

Expected answer / marking points: Arc $= 60/360 \times 2 \times 22/7 \times 21 = 22$ cm [1.5]; area $= 60/360 \times 22/7 \times 441 = 231$ cm^2 [1.5].

Q29 **[3]** **Question:** Solve: $0.2x + 0.3y = 1.3$ and $0.4x + 0.5y = 2.3$.

Expected answer / marking points: $2x + 3y = 13$, $4x + 5y = 23$ [1]; double first and subtract: $y = 3$ [1]; $x = 2$ [1].

Q30 **[3]** **Question:** Vyom's marks and subject credits are: Maths 80 (4 credits), Science 70 (3), English 90 (2), SST 60 (3). Find his credit-weighted average. Is it more or less than his simple average?

Expected answer / marking points: Weighted $= (320 + 210 + 180 + 180)/12 = 890/12 \approx 74.17$ [2]; simple $= 300/4 = 75$; weighted is slightly less [1].

Q31 **[3]** **Question:** ABCD is a parallelogram. AP and CQ are perpendiculars from A and C to the diagonal BD. Show that AP = CQ.

Expected answer / marking points: $\triangle APB$ and $\triangle CQD$: $\angle APB = \angle CQD = 90°$ [1]; $\angle ABP = \angle CDQ$ (alternate, $AB \parallel DC$) and $AB = CD$ [1]; AAS $\Rightarrow$ congruent $\Rightarrow$ $AP = CQ$ [1].

Q32 **[5]** **Question:** Solve graphically: $x + 2y = 6$ and $2x - y = 2$. Also find the area of the triangle formed by the two lines and the x-axis.

ATHVA / OR

A father is three times as old as his son. After 10 years, he will be twice as old as the son. Form the equations, solve them, and check your answer.

Expected answer / marking points: Tables + graph [2]; solution (2, 2) [1]; x-intercepts (6, 0) and (1, 0) $\Rightarrow$ base 5 [1]; area $= \frac{1}{2} \times 5 \times 2 = 5$ sq units [1].
OR: $f = 3s$ and $f + 10 = 2(s + 10)$ [2]; $3s + 10 = 2s + 20 \Rightarrow s = 10$, $f = 30$ [2]; check: after 10 years $40 = 2 \times 20$ ✓ [1].

Q33 [5] **Question:** A ticket is drawn at random from tickets numbered 1 to 30. Find the probability that the number is (i) even (ii) a multiple of 3 (iii) a prime (iv) a perfect square (v) a multiple of both 3 and 5.

Expected answer / marking points: (i) 15/30 = 1/2 [1] (ii) 10/30 = 1/3 [1] (iii) 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 ⇒ 10/30 = 1/3 [1] (iv) 1, 4, 9, 16, 25 ⇒ 5/30 = 1/6 [1] (v) 15, 30 ⇒ 2/30 = 1/15 [1].

Q34 [5] **Question:** A ball is dropped from a height of 64 m. After each bounce it rises to half the height of the previous fall. (i) Write the heights of the first four bounces. (ii) Write a formula for the height of the n-th bounce. (iii) After which bounce will the height first be less than 1 m? (iv) Find the total distance travelled by the ball until it touches the ground for the 4th time.

Expected answer / marking points: (i) 32, 16, 8, 4 m [1]. (ii) $h_n = 64 \times (½)^n$ [1]. (iii) $64/2^n < 1 \Rightarrow 2^n > 64 \Rightarrow n = 7$ (0.5 m) [1.5]. (iv) 64 + 2(32 + 16 + 8) = 176 m [1.5].

Q35 [5] **Question:** Water flows through a cylindrical pipe of inner diameter 7 cm at a speed of 5 m per second. How many litres of water flow out in 10 minutes?

ATHVA / OR

A conical tent has base radius 7 m and height 24 m. Find (i) its slant height (ii) the canvas needed and its cost at ₹70 per m^2 (iii) the volume of air inside.

Expected answer / marking points: In 1 s, water forms a cylinder of r = 3.5 cm, h = 500 cm [1]; volume = 22/7 × 12.25 × 500 = 19,250 cm^3 = 19.25 L [2]; in 600 s: 11,550 L [2].
OR: (i) l = √(49 + 576) = 25 m [1]. (ii) πrl = 22/7 × 7 × 25 = 550 m^2; cost ₹38,500 [2]. (iii) ⅓ × 22/7 × 49 × 24 = 1232 m^3 [2].

Q36 [4] **Question:**

> A classroom seating plan is drawn on a grid. The teacher's desk is at the origin. Raj sits at (2, 3) and Meena at (2, 7).

(i) How far apart are Raj and Meena? [1]
(ii) Arun sits on the y-axis, level with Raj. Write his coordinates. [1]
(iii) The cupboard is 3 units to the left of the y-axis and 2 units below the x-axis. Write its coordinates and quadrant, and find its distance from a point (–3, 5). [2]

Expected answer / marking points: (i) |7 – 3| = 4 units [1]. (ii) (0, 3) [1]. (iii) (–3, –2), quadrant III [1]; distance |5 – (–2)| = 7 units [1].

Q37 [4] **Question:**

> A sapling is 5 cm tall when planted and grows 2 cm every week.

(i) What will its height be after 6 weeks? [1]
(ii) Write the height h after w weeks as a linear polynomial. What does the slope represent? [1]
(iii) After how many weeks will it be 35 cm tall? If another plant 20 cm tall shrinks by 1 cm every week, write its rule and state whether this is linear growth or decay. [2]

Expected answer / marking points: (i) 17 cm [1]. (ii) h = 2w + 5; slope = growth per week [1]. (iii) 2w + 5 = 35 ⇒ w = 15 weeks [1]; h = 20 – w, linear decay (slope –1) [1].

Q38 **Question:**

[4]

> In Class 9A, 18 of 30 students like cricket. In Class 9B, 12 of 20 students like cricket.

(i) What percentage of 9A likes cricket? [1]
(ii) What percentage of 9B likes cricket? [1]
(iii) Find the percentage for both classes together. If instead 8 of the 20 students in 9B liked cricket, would the combined percentage be the simple average of 60% and 40%? Explain. [2]

Expected answer / marking points: (i) 60% [1]. (ii) 60% [1]. (iii) 30/50 = 60% [1]; no — combined = 26/50 = 52%, not 50%, because the classes have different sizes (weighted average) [1].

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

Science · Mock Paper 2 · Answer key

Tuesday, 23 February 2027 · 80 marks

Test ID: T2027-02-23

Class 9 · Exploration — poora syllabus (Ch 1–13)

Q1 **[1]**

Question: A variable that is deliberately changed in an experiment is the:

(a) controlled variable (b) independent variable

(c) dependent variable (d) constant

Expected answer / marking points: (b) independent variable

Q2 **[1]**

Question: The cell organelle that packages and dispatches materials is the:

(a) Golgi apparatus (b) lysosome

(c) ribosome (d) vacuole

Expected answer / marking points: (a) Golgi apparatus

Q3 **[1]**

Question: Bones are joined to other bones by:

(a) tendons (b) ligaments

(c) cartilage only (d) muscles

Expected answer / marking points: (b) ligaments

Q4 **[1]**

Question: The slope of a distance–time graph gives:

(a) acceleration (b) speed

(c) displacement (d) force

Expected answer / marking points: (b) speed

Q5 **[1]**

Question: Fog is an example of a colloid in which:

(a) solid is dispersed in gas (b) liquid is dispersed in gas

(c) gas is dispersed in liquid (d) liquid is dispersed in solid

Expected answer / marking points: (b) liquid is dispersed in gas

Q6 **[1]**

Question: A force of 10 N acts on a body for 2 s. The change in its momentum is:

(a) 5 kg m s^{-1} (b) 20 kg m s^{-1}

(c) 12 kg m s^{-1} (d) 8 kg m s^{-1}

Expected answer / marking points: (b) 20 kg m s^{-1}

Q7 **[1]**

Question: When the speed of a body is doubled, its kinetic energy becomes:

(a) double (b) half

(c) four times (d) same

Expected answer / marking points: (c) four times

Q8 [1]

Question: The maximum number of electrons in the second shell is:

(a) 2 (b) 8

(c) 18 (d) 32

Expected answer / marking points: (b) 8

Q9 [1]

Question: Atoms of the same element with different mass numbers are called:

(a) isobars (b) isotopes

(c) ions (d) molecules

Expected answer / marking points: (b) isotopes

Q10 [1]

Question: The molecular mass of O_2 is (O = 16 u):

(a) 16 u (b) 32 u

(c) 8 u (d) 48 u

Expected answer / marking points: (b) 32 u

Q11 [1]

Question: Sound travels fastest in:

(a) vacuum (b) air

(c) water (d) iron

Expected answer / marking points: (d) iron

Q12 [1]

Question: A sound of frequency 15 Hz is called:

(a) ultrasonic (b) audible

(c) infrasonic (d) supersonic

Expected answer / marking points: (c) infrasonic

Q13 [1]

Question: After fertilisation, the ovary of a flower develops into the:

(a) seed (b) fruit

(c) embryo (d) petal

Expected answer / marking points: (b) fruit

Q14 [1]

Question: Which of these is a protist?

(a) Mushroom (b) Bacteria

(c) Paramecium (d) Fern

Expected answer / marking points: (c) Paramecium

Q15 [1]

Question: Which of the following has the highest albedo?

(a) Fresh snow (b) Forest

(c) Ocean (d) Asphalt road

Expected answer / marking points: (a) Fresh snow

Q16 **Question:** The gas released by plants during photosynthesis is:
[1]

(a) carbon dioxide (b) nitrogen

(c) oxygen (d) methane

Expected answer / marking points: (c) oxygen

Q17 **Question: A:** Atoms combine to complete their outermost shells.
[1] **R:** Atoms with 8 electrons (or 2 for the first shell) in the outermost shell are stable.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q18 **Question: A:** A ball thrown vertically up has zero velocity at the highest point but its acceleration is not zero.
[1] **R:** Gravity acts on the ball at all times.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q19 **Question: A:** Reverberation is desirable in large concert halls in very large amounts.
[1] **R:** Excessive reverberation makes speech and music unclear.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (d)

Q20 **Question: A:** Fossils help scientists understand how life has changed over time.
[1] **R:** Fossils are preserved remains or traces of organisms from the past.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q21 **Question:** Write two functions of the cell membrane.
[2]

Expected answer / marking points: Controls entry and exit of substances (selectively permeable) [1]; gives shape and protects the cell [1].

Q22 **Question:** Draw velocity–time graphs for (i) uniform velocity (ii) uniform acceleration from rest.
[2]

Expected answer / marking points: (i) Horizontal line parallel to the time axis [1] (ii) Straight line through the origin with positive slope [1].

Q23 **Question:** What is a saturated solution? What happens when a hot saturated solution is cooled?
[2]

Expected answer / marking points: A solution that cannot dissolve more solute at that temperature [1]; on cooling, extra solute crystallises out [1].

Q24 **Question:** A man of mass 60 kg climbs to a height of 10 m in 20 s. Find his power.
[2]

Expected answer / marking points: W = 60 × 10 × 10 = 6000 J [1]; P = 300 W [1].

Q25 **Question:** Why does sexual reproduction need meiosis?
[2]

Expected answer / marking points: Gametes must have half the chromosomes [1]; so that the zygote formed after fusion has the normal number [1].

Q26 **Question:** How does deforestation affect the carbon cycle?
[2]

Expected answer / marking points: Fewer trees remove less CO_2 by photosynthesis [1]; burning/decay of cut trees releases more CO_2 → global warming [1].

Q27 **Question:** Distinguish between parenchyma, collenchyma and sclerenchyma.
[3]

Expected answer / marking points: Parenchyma: thin walls, loosely packed, storage [1]; collenchyma: thickened corners, flexibility (leaf stalks) [1]; sclerenchyma: thick lignified dead cells, strength (husk, seed coat) [1].

Q28 **Question:** A stone is thrown vertically upward with a speed of 20 m s^{-1}. Find (i) the maximum height (ii) the time to reach it (iii) the total time before it returns to the thrower's hand.
[3]

Expected answer / marking points: (i) $v^2 = u^2 - 2gh \Rightarrow h = 400/20 = 20$ m [1] (ii) $t = 20/10 = 2$ s [1] (iii) 4 s [1].

Q29 **Question:** Differentiate between homogeneous and heterogeneous mixtures with two examples each. Is air a mixture or a compound?
[3]

Expected answer / marking points: Homogeneous — uniform composition (salt solution, air, brass) [1]; heterogeneous — non-uniform (sand + water, oil + water) [1]; air is a mixture (variable composition, components keep their properties) [1].

Q30 **Question:** Two forces of 5 N and 4 N act on a block in opposite directions on a smooth surface. Another block moves at constant velocity. Which experiences a net force? Explain using the first law.
[3]

Expected answer / marking points: First block: net 1 N towards the 5 N force — it accelerates [1.5]; second: constant velocity ⇒ net force zero (first law) [1.5].

Q31 **Question:** State the law of constant proportions. In water, hydrogen and oxygen combine in a mass ratio of 1 : 8. How much oxygen combines with 3 g of hydrogen, and how much water forms?
[3]

Expected answer / marking points: Statement [1]; oxygen = 24 g [1]; water = 27 g [1].

Q32 [3] **Question:** Explain how the human ear distinguishes loud and soft sounds and high and low pitched sounds, with reference to amplitude and frequency.

Expected answer / marking points: Loudness depends on amplitude — larger amplitude, louder [1.5]; pitch depends on frequency — higher frequency, higher pitch (e.g., whistle vs drum) [1.5].

Q33 [3] **Question:** What are the parts of a flower involved in reproduction? Describe what happens after a pollen grain lands on the stigma.

Expected answer / marking points: Stamen and carpel (stigma, style, ovary) [1]; pollen tube grows through style to ovule [1]; male gamete fuses with egg → zygote → embryo; ovule → seed, ovary → fruit [1].

Q34 [5] **Question:** State and explain the law of conservation of mechanical energy with the example of a freely falling body. A 1 kg ball falls from 10 m. Make a table of PE, KE and total energy at 10 m, 5 m and 0 m.

ATHVA / OR

Define work. When is work positive, negative or zero? Give one example of each. A force of 15 N moves a trolley 8 m in the direction of the force. Find the work done.

Expected answer / marking points: Statement + explanation [2]; table: 10 m — 100, 0, 100 J; 5 m — 50, 50, 100 J; 0 m — 0, 100, 100 J [3].
OR: Definition (W = F × s in the direction of force) [1]; positive (lifting a bag), negative (friction on a sliding box), zero (carrying a bag on level ground / force ⊥ displacement) [3]; W = 120 J [1].

Q35 [5] **Question:** Explain the formation of (i) sodium chloride (ii) a water molecule. Write their formulae, the type of bond and calculate their formula unit/molecular masses (Na = 23, Cl = 35.5, H = 1, O = 16).

ATHVA / OR

Describe the human male and female reproductive systems briefly. Why is it important to maintain hygiene and good health during adolescence?

Expected answer / marking points: NaCl: Na gives 1 e^- to Cl → ionic; 58.5 u [2.5]; H_2O: O shares one electron pair with each of two H → covalent; 18 u [2.5].
OR: Male: testes produce sperms and hormones; sperm duct, accessory glands [1.5]; female: ovaries produce eggs and hormones; oviduct (fertilisation), uterus (development of embryo) [2]; hygiene and nutrition prevent infections and support healthy growth [1.5].

Q36 [5] **Question:** Explain the water cycle and the nitrogen cycle with labelled flow diagrams. Why are these cycles important for life?

Expected answer / marking points: Water cycle — evaporation, transpiration, condensation, precipitation, run-off, infiltration [2]; nitrogen cycle — fixation, nitrification, assimilation, ammonification/decomposition, denitrification [2]; recycle essential water and nutrients for all organisms [1].

Q37 **Question:**

[4]

> A car on a straight road starts from rest, accelerates uniformly to 20 m s^{-1} in 10 s, moves at this speed for 20 s, and then stops uniformly in 5 s.

(i) Find the acceleration in the first 10 s. [1]
(ii) Find the retardation in the last 5 s. [1]
(iii) Draw the v–t graph and find the total distance travelled. [2]

Expected answer / marking points: (i) 2 m s^{-2} [1]. (ii) 4 m s^{-2} [1]. (iii) Graph (trapezium) [1]; distance = 100 + 400 + 50 = 550 m [1].

Q38 **Question:**

[4]

> A doctor uses an ultrasound scanner to see a baby inside the mother's womb. The machine sends sound of 2 MHz into the body and records the echoes from different organs.

(i) What is ultrasound? [1]
(ii) Why is ultrasound used instead of X-rays for this? [1]
(iii) Give two other uses of ultrasound and explain how echoes help form an image. [2]

Expected answer / marking points: (i) Sound above 20 kHz [1]. (ii) It is safe — no harmful radiation [1]. (iii) Cleaning small parts, detecting cracks in metal, SONAR, echocardiography (any two) [1]; different tissues reflect sound differently; time and strength of echoes are used to build the image [1].

Q39 **Question:**

[4]

> A student found an organism under a microscope: single-celled, with a well-defined nucleus and many cilia. In the garden she also found an animal with a segmented body and jointed legs.

(i) To which kingdom does the microscopic organism belong? [1]
(ii) Which feature tells you the garden animal is an insect/arthropod and not an earthworm? [1]
(iii) Why is cellular organisation (prokaryotic/eukaryotic, unicellular/multicellular) a basic feature for classification? [2]

Expected answer / marking points: (i) Protista [1]. (ii) Jointed legs [1]. (iii) It is a fundamental property that decides many other features [1]; it separates major groups like Monera, Protista and multicellular kingdoms clearly [1].

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

Social Science · Mock Paper 2 · Answer key

Wednesday, 24 February 2027 · 80 marks

Test ID: T2027-02-24

Class 9 · Part 1 (Ch 1–9) + Part 2 (Ch 1–6) — poora syllabus

Q1 [1]

Question: Social Science helps us mainly to:

(a) memorise dates
(b) understand society, its past and present
(c) study only maps
(d) learn only about money

Expected answer / marking points: (b) understand society, its past and present

Q2 [1]

Question: The breaking and decay of rocks in place is called:

(a) erosion
(b) weathering
(c) deposition
(d) faulting

Expected answer / marking points: (b) weathering

Q3 [1]

Question: Moraines are deposits made by:

(a) wind
(b) glaciers
(c) rivers
(d) waves

Expected answer / marking points: (b) glaciers

Q4 [1]

Question: Average weather of a place over a long period is called:

(a) weather
(b) climate
(c) season
(d) humidity

Expected answer / marking points: (b) climate

Q5 [1]

Question: Early humans are believed to have evolved first in:

(a) Asia
(b) Europe
(c) Africa
(d) Australia

Expected answer / marking points: (c) Africa

Q6 [1]

Question: Which civilisation developed along the Nile?

(a) Mesopotamian
(b) Egyptian
(c) Chinese
(d) Sindhu–Sarasvatī

Expected answer / marking points: (b) Egyptian

Q7 [1]

Question: Guilds of craftsmen and merchants in early India were called:

(a) janas
(b) śreṇīs
(c) samitis
(d) gaṇas

Expected answer / marking points: (b) śreṇīs

Q8 [1] **Question:** The Constitution of India gives citizens:

(a) only duties (b) only rights

(c) fundamental rights and duties (d) no rights

Expected answer / marking points: (c) fundamental rights and duties

Q9 [1] **Question:** VVPAT is used along with the EVM to:

(a) count votes faster (b) let voters verify their vote on a paper slip

(c) register new voters (d) announce results

Expected answer / marking points: (b) let voters verify their vote on a paper slip

Q10 [1] **Question:** Which of these is a planned (command) economy feature?

(a) Prices set by demand and supply (b) Government decides what to produce

(c) Only private firms (d) No government at all

Expected answer / marking points: (b) Government decides what to produce

Q11 [1] **Question:** When price rises, other things being equal, quantity supplied:

(a) falls (b) rises

(c) remains constant (d) becomes zero

Expected answer / marking points: (b) rises

Q12 [1] **Question:** The high seas are:

(a) under one country's control (b) open to all countries

(c) only for fishing (d) part of the EEZ

Expected answer / marking points: (b) open to all countries

Q13 [1] **Question:** Which of these is a cause of forest fires?

(a) Lightning and careless burning (b) Rainfall

(c) Snowfall (d) Tides

Expected answer / marking points: (a) Lightning and careless burning

Q14 [1] **Question:** The Battle of Talikota (1565) weakened the:

(a) Mughal Empire (b) Vijayanagara Empire

(c) Ahom kingdom (d) Maratha state

Expected answer / marking points: (b) Vijayanagara Empire

Q15 [1] **Question:** The Bhakti saints spread their message mostly through:

(a) Persian only (b) local languages

(c) Latin (d) English

Expected answer / marking points: (b) local languages

Q16 [1]

Question: Which of these is a principle of good governance?

(a) Accountability
(b) Secrecy
(c) Favouritism
(d) Delay

Expected answer / marking points: (a) Accountability

Q17 [1]

Question: A budget helps a family to:

(a) spend without limit
(b) plan expenses and save
(c) avoid paying taxes
(d) stop earning

Expected answer / marking points: (b) plan expenses and save

Q18 [1]

Question: Investments with higher possible returns usually carry:

(a) no risk
(b) lower risk
(c) higher risk
(d) guaranteed profit

Expected answer / marking points: (c) higher risk

Q19 [1]

Question: The money coming in and going out of a business over time is its:

(a) profit
(b) cash flow
(c) capital
(d) loan

Expected answer / marking points: (b) cash flow

Q20 [1]

Question: Startups contribute to the economy mainly by:

(a) reducing jobs
(b) creating jobs and innovation
(c) increasing imports only
(d) avoiding taxes

Expected answer / marking points: (b) creating jobs and innovation

Q21 [2]

Question: Why do we not feel the pressure of the atmosphere?

Expected answer / marking points: The air pressure acts equally from all sides [1]; our body's internal pressure balances it [1].

Q22 [2]

Question: Name any two features of Bronze Age civilisations.

Expected answer / marking points: Urban centres/cities, trade, writing, administration, social organisation, metal (bronze) tools (any two) [2].

Q23 [2]

Question: What is meant by "rule of law"?

Expected answer / marking points: Everyone, including rulers and officials, is equal before the law [1]; no one is above the law; laws apply equally [1].

Q24 [2]

Question: Give two reasons why oceans are important for climate.

Expected answer / marking points: Absorb and store heat, moderate temperatures [1]; currents carry heat; source of moisture for rain/monsoon [1].

Q25 **Question:** How do running water and wind shape the Earth's surface? Give one landform of each.
[3]

Expected answer / marking points: Running water erodes valleys and deposits sediment — V-shaped valley, meander, delta, waterfall [1.5]; wind erodes and deposits in deserts — sand dunes, mushroom rocks [1.5].

Q26 **Question:** How did political organisation change from the Vedic period to the age of empires?
[3]

Expected answer / marking points: Clans (janas) led by a rājā with sabhā and samiti [1]; larger kingdoms (janapadas, mahājanapadas) and republics (gaṇas) [1]; empires like the Mauryas and Guptas with structured administration, taxes and armies [1].

Q27 **Question:** Explain three ways in which citizens can strengthen democracy in everyday life.
[3]

Expected answer / marking points: Voting and staying informed [1]; responsible use of media — checking facts before sharing [1]; taking part in community activities, respecting others' views and laws [1].

Q28 **Question:** Explain the law of supply and two factors other than price that affect supply.
[3]

Expected answer / marking points: Law: other things same, quantity supplied rises with price [1]; technology — better technology lowers cost, raises supply [1]; input costs/weather/number of sellers (any one) [1].

Q29 **Question:** Why are forests important for communities living in and around them? Give three points.
[3]

Expected answer / marking points: Food, fuelwood, fodder and medicinal plants [1]; livelihoods from forest produce and ecotourism [1]; cultural/spiritual importance and protection of soil and water [1].

Q30 **Question:** Explain the movements of ocean water — waves, tides and currents — and their effects on
[5] human activities.

ATHVA / OR

What are maritime zones? Explain territorial waters, the EEZ and the high seas, and how UNCLOS helps settle disputes.

Expected answer / marking points: Waves — caused by wind; erosion, surfing, wave energy [1.5]; tides — caused by Moon/Sun; help navigation, fishing, tidal energy [1.5]; currents — warm/cold; affect climate, fishing grounds, shipping routes [2].
OR: Territorial sea up to 12 nm — full sovereignty, innocent passage allowed [1.5]; EEZ up to 200 nm — rights over resources [1.5]; high seas — open to all [1]; UNCLOS defines zones and provides rules/tribunals to settle disputes [1].

Q31 **Question:** Why is India called the world's largest democracy? What challenges does it face, and how can
[5] they be met?

Expected answer / marking points: Largest number of registered voters and scale of participation [1.5]; challenges — unequal participation, misinformation, money/muscle power, inequality [2]; solutions — awareness, ECI reforms, free press, civic education [1.5].

Q32 [5] **Question:** What is opportunity cost? Explain with three examples from daily life and show how it helps in making better decisions.

ATHVA / OR

Explain market equilibrium with a demand–supply schedule and diagram. What happens to the equilibrium price when demand increases?

Expected answer / marking points: Definition [1]; three examples (e.g., studying vs playing, saving vs spending, farmer choosing crops) [3]; helps compare choices and use scarce resources wisely [1].
OR: Schedule showing Qd = Qs at one price [1.5]; diagram [1.5]; demand rises ⇒ shortage at old price ⇒ price and quantity rise to a new equilibrium [2].

Q33 [5] **Question:** Explain the five pillars of personal finance. How do saving and investing differ?

Expected answer / marking points: Income, budgeting, saving, investing, protection — each explained [3.5]; saving keeps money aside safely (low return); investing puts money to work for higher returns with some risk [1.5].

Q34 [4] **Question:**

> An elephant herd raided crops in a village next to a forest. Villagers say the elephants' old path to a water hole was blocked by a new resort and fencing. The forest department proposed a corridor and solar-powered warning alarms.

(i) What is this problem called? [1]
(ii) What is the main cause in this case? [1]
(iii) Suggest two ways to reduce such conflicts and protect both people and wildlife. [2]

Expected answer / marking points: (i) Human–wildlife conflict [1]. (ii) Loss/blocking of the elephants' habitat and path (corridor) [1]. (iii) Wildlife corridors, early-warning systems, compensation for crop loss, community awareness, no construction on animal routes (any two) [2].

Q35 [4] **Question:**

> A ward committee wants to improve garbage collection. It holds a public meeting, publishes the collection timetable online, sets up a complaint number, and asks residents to separate wet and dry waste.

(i) Which level of governance is involved here? [1]
(ii) Which principle is shown by publishing the timetable? [1]
(iii) What is the role of citizens in making this policy successful? Name two other actors who can help. [2]

Expected answer / marking points: (i) Local governance (municipal/ward) [1]. (ii) Transparency [1]. (iii) Citizens separate waste, follow timings, give feedback [1]; NGOs/resident welfare associations, private waste agencies, media (any two) [1].

Q36 **Question:**
[4]

> Priya, 15, wants to start a weekend stall selling handmade bookmarks. She expects to spend ₹600 on material and earn ₹1,500 in the first month. Her uncle advises her to write a simple business plan first.

(i) What will be her expected profit in the first month? [1]
(ii) Name one risk she might face. [1]
(iii) What two things should her simple business plan include, and why? [2]

Expected answer / marking points: (i) ₹900 [1]. (ii) Low sales, competition, rain, damaged stock (any one) [1]. (iii) Any two: goal and product, target customers, costs and prices, marketing plan, record keeping — helps her organise and decide [2].

Q37 **Question:** On an outline political map of India, locate and label any five: (a) the Thar desert (b) the Ganga
[5] river (c) Lakshadweep (d) the Eastern Ghats (e) Hampi (f) Udaipur (Mewar) (g) a state with a long coastline on the Arabian Sea.

Expected answer / marking points: Each correctly located and labelled [1 × 5]; (g) e.g., Gujarat/Maharashtra/Kerala.

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

हिंदी · मॉक प्रश्न-पत्र 2 · Answer key

Thursday, 25 February 2027 · 80 marks

Test ID: T2027-02-25

कक्षा 9 · गंगा (पाठ 1–12) + व्याकरण + लेखन — पूरा पाठ्यक्रम

Q1 **[7]**

Question: निम्नलिखित गद्यांश पढ़कर प्रश्नों के उत्तर दीजिए —

> खेल केवल मनोरंजन का साधन नहीं हैं; वे जीवन की पाठशाला भी हैं। मैदान पर बच्चा हारना और फिर से उठकर खड़ा होना सीखता है। टीम के साथ खेलते हुए वह समझता है कि अकेले की जीत से बड़ी सबकी जीत होती है।
> आजकल बहुत-से बच्चे खेल के मैदान की जगह मोबाइल की स्क्रीन पर समय बिताते हैं। इससे आँखों पर ज़ोर पड़ता है, नींद कम होती है और शरीर सुस्त हो जाता है। डॉक्टरों के अनुसार किशोरों को प्रतिदिन कम-से-कम एक घंटा शारीरिक गतिविधि करनी चाहिए। तैराकी, दौड़, साइकिल चलाना या कोई भी खेल — शरीर के साथ-साथ मन को भी स्वस्थ रखता है। जो विद्यार्थी पढ़ाई और खेल में संतुलन रखते हैं, वे परीक्षा के तनाव को भी बेहतर ढंग से सँभाल पाते हैं।

(i) खेलों को “जीवन की पाठशाला” क्यों कहा गया है? [2]
(ii) मोबाइल पर अधिक समय बिताने के दो नुकसान लिखिए। [1]
(iii) डॉक्टरों के अनुसार किशोरों को प्रतिदिन कितनी शारीरिक गतिविधि करनी चाहिए? [1]
(iv) पढ़ाई और खेल में संतुलन रखने का क्या लाभ है? [1]
(v) “सुस्त” का विलोम शब्द लिखिए। [1]
(vi) गद्यांश का उपयुक्त शीर्षक लिखिए। [1]

Expected answer / marking points: (i) बच्चा हारकर फिर उठना और टीम के साथ मिलकर जीतना सीखता है [2]। (ii) आँखों पर ज़ोर, कम नींद, शरीर सुस्त (कोई दो) [1]। (iii) कम-से-कम एक घंटा [1]। (iv) परीक्षा का तनाव बेहतर ढंग से सँभलता है [1]। (v) चुस्त/फुर्तीला [1]। (vi) “खेल — जीवन की पाठशाला” (कोई उपयुक्त) [1]।

Q2 **[7]**

Question: निम्नलिखित काव्यांश पढ़कर प्रश्नों के उत्तर दीजिए —

> नन्ही चिड़िया डाल पे बैठी, तिनका-तिनका जोड़ रही,
> आँधी आई, घोंसला टूटा, फिर भी हिम्मत छोड़ नहीं।
> फिर से उड़ी वह दूर गगन में, फिर से तिनके ले आई,
> मेहनत की उस नन्ही जान ने, सबको यह बात सिखलाई —
> गिरकर उठना, उठकर चलना, यही है जीवन की रीत,
> जो हार न माने मुश्किल से, उसकी ही होती है जीत।

(i) चिड़िया क्या कर रही थी? [1]
(ii) आँधी आने पर क्या हुआ? [1]
(iii) घोंसला टूटने पर चिड़िया ने क्या किया? [1]
(iv) “जीवन की रीत” क्या बताई गई है? [1]
(v) “गगन” का पर्यायवाची लिखिए। [1]
(vi) काव्यांश से मिलने वाली सीख लिखिए। [2]

Expected answer / marking points: (i) तिनका-तिनका जोड़कर घोंसला बना रही थी [1]। (ii) घोंसला टूट गया [1]। (iii) हिम्मत नहीं हारी, फिर से तिनके ले आई [1]। (iv) गिरकर उठना और उठकर चलना [1]। (v) आकाश/आसमान/नभ [1]। (vi) मुश्किलों से हार न मानकर मेहनत करते रहने वाले की ही जीत होती है [2]।

Q3 **[2]**

Question: उपसर्ग लगाकर दो-दो शब्द बनाइए — (क) “वि” (ख) “अधि”

Expected answer / marking points: (क) विज्ञान, विदेश, वियोग (कोई दो) [1] (ख) अधिकार, अधिकारी, अधिपति (कोई दो) [1]।

Q4 [2]

Question: प्रत्यय लगाकर शब्द बनाइए — (क) पढ़ + आई (ख) धन + वान

Expected answer / marking points: (क) पढ़ाई [1] (ख) धनवान [1]।

Q5 [2]

Question: अर्थ के आधार पर वाक्य-भेद लिखिए — (क) राम आज नहीं आएगा। (ख) भगवान तुम्हें सफलता दे।

Expected answer / marking points: (क) निषेधवाचक वाक्य [1] (ख) इच्छावाचक वाक्य [1]।

Q6 [2]

Question: निर्देशानुसार वाक्य बदलिए — (क) “वह बीमार होने के कारण विद्यालय नहीं गया।” (संयुक्त वाक्य में) (ख) “जो लड़का कक्षा में प्रथम आया, वह मेरा भाई है।” (सरल वाक्य में)

Expected answer / marking points: (क) वह बीमार था, इसलिए विद्यालय नहीं गया [1] (ख) कक्षा में प्रथम आने वाला लड़का मेरा भाई है [1]।

Q7 [2]

Question: अलंकार पहचानिए — (क) “पायो जी मैंने राम रतन धन पायो।” (ख) “तरनि-तनूजा तट तमाल तरुवर बहु छाए।”

Expected answer / marking points: (क) रूपक — राम-नाम को रत्न-धन कहा [1] (ख) अनुप्रास — “त” वर्ण की आवृत्ति [1]।

Q8 [2]

Question: वाच्य बदलिए — (क) माली पौधों को पानी देता है। (ख) चलो, अब सोया जाए।

Expected answer / marking points: (क) माली द्वारा पौधों को पानी दिया जाता है [1] (ख) चलो, अब सोएँ [1]।

Q9 [2]

Question: रेखांकित पदों का परिचय दीजिए — “सुभद्रा जी ने वीरतापूर्ण कविता लिखी।”

Expected answer / marking points: सुभद्रा जी — व्यक्तिवाचक संज्ञा, स्त्रीलिंग, एकवचन, कर्ता [1]; वीरतापूर्ण — गुणवाचक विशेषण, “कविता” विशेष्य [1]।

Q10 [2]

Question: मुहावरों का अर्थ लिखकर वाक्य में प्रयोग कीजिए — (क) हिम्मत न हारना (ख) आँखें खुलना

Expected answer / marking points: (क) साहस बनाए रखना — वाक्य [1] (ख) सच्चाई का पता चलना/सावधान होना — वाक्य [1]।

Q11 [5]

Question: पाठ 6 “रीढ़ की हड्डी” पाठ्यपुस्तक में खोलिए और एकांकी का अंतिम दृश्य पढ़िए जहाँ उमा गोपाल प्रसाद से बात करती है। फिर उत्तर दीजिए —
(क) उमा ने अपनी पढ़ाई के बारे में क्या सच बताया? [1]
(ख) उमा ने गोपाल प्रसाद से क्या प्रश्न किए? [2]
(ग) उमा के इस साहस से हमें क्या सीख मिलती है? [2]

Expected answer / marking points: (क) कि वह बी.ए. तक पढ़ी है (पाठ के अनुसार) [1]। (ख) क्या लड़कियाँ बिकाऊ वस्तु हैं जिन्हें परखा जाए; आपका बेटा पढ़ाई के समय कैसे चरित्र का था (पाठ के अनुसार भाव) [2]। (ग) अन्याय और अपमान के विरुद्ध आवाज़ उठानी चाहिए; लड़कियों की शिक्षा और सम्मान आवश्यक है [2]।

Q12 [5]

Question: निम्नलिखित पंक्तियाँ पढ़कर प्रश्नों के उत्तर दीजिए —

> बहु धनुही तोरी लरिकाईं। कबहुँ न असि रिस कीन्हि गोसाईं॥
> येहि धनु पर ममता केहि हेतू। सुनि रिसाइ कह भृगुकुलकेतू॥

(क) ये पंक्तियाँ कौन किससे कह रहा है? [1]
(ख) लक्ष्मण ने बचपन की किस बात की याद दिलाई? [1]
(ग) “भृगुकुलकेतू” किसे कहा गया है? [1]
(घ) इन पंक्तियों में लक्ष्मण के स्वभाव की कौन-सी विशेषता दिखती है? [2]

Expected answer / marking points: (क) लक्ष्मण परशुराम से [1]। (ख) बचपन में उन्होंने बहुत-सी धनुहियाँ तोड़ीं, तब परशुराम ने कभी ऐसा क्रोध नहीं किया [1]। (ग) परशुराम को (भृगुवंश के ध्वज/श्रेष्ठ) [1]। (घ) व्यंग्यपूर्ण, निडर और तर्कशील स्वभाव [2]।

Q13 [2]

Question: “ऐसी भी बातें होती हैं” पाठ में लता मंगेशकर की संगीत-साधना के बारे में दो बातें लिखिए।

Expected answer / marking points: पाँच वर्ष की आयु में पिता से संगीत सीखना शुरू किया [1]; जीवनभर संगीत के प्रति समर्पित रहीं, परिवार की ज़िम्मेदारी भी निभाई [1]।

Q14 [2]

Question: “दो बैलों की कथा” में झूरी की पत्नी हीरा-मोती से क्यों नाराज़ थी?

Expected answer / marking points: बैल गया के घर से भागकर लौट आए थे [1]; उसे लगा कि वे काम से जी चुराते हैं, इसलिए उन्हें सूखा भूसा दिलवाया [1]।

Q15 [2]

Question: “मैं और मेरा देश” निबंध का मुख्य संदेश क्या है?

Expected answer / marking points: व्यक्ति का सम्मान देश के सम्मान से जुड़ा है [1]; हर नागरिक को अपने आचरण से देश की शक्ति और सुंदरता बढ़ानी चाहिए [1]।

Q16 [2]

Question: रैदास ने अपने प्रभु को “गरीब निवाजु” क्यों कहा है?

Expected answer / marking points: प्रभु गरीबों और दीनों पर कृपा करते हैं [1]; उन्होंने छोटी समझी जाने वाली जाति के भक्त को भी ऊँचा सम्मान दिया [1]।

Q17 [2]

Question: “झाँसी की रानी” कविता में अंग्रेज़ों की किस नीति की ओर संकेत है?

Expected answer / marking points: राज्य हड़पने की नीति (डलहौज़ी की हड़प नीति) [1]; दत्तक पुत्र को उत्तराधिकारी न मानकर झाँसी पर कब्जा [1]।

Q18 [2]

Question: “घर की याद” कविता में कवि माँ के बारे में क्या कहते हैं?

Expected answer / marking points: माँ पढ़ी-लिखी नहीं पर ममतामयी हैं [1]; बेटे की याद में दुखी होंगी, कवि उनकी चिंता करते हैं (पाठ के अनुसार भाव) [1]।

Q19 [4]

Question: “दो बैलों की कथा” के आधार पर सिद्ध कीजिए कि हीरा और मोती में सच्ची मित्रता थी। (60–80 शब्द)

Expected answer / marking points: हर संकट में साथ रहे — गया के घर, काँजीहौस, साँड़ से लड़ाई [2]; एक-दूसरे के लिए जोखिम उठाया, मोती हीरा को छोड़कर नहीं भागा [1.5]; निष्कर्ष [0.5]।

Q20 [4]

Question: “आखिरी चट्टान तक” यात्रा-वृत्तांत पढ़कर आपके मन में कन्याकुमारी की कौन-सी छवि बनती है? (60–80 शब्द)

Expected answer / marking points: भारत का दक्षिणी छोर, तीन समुद्रों का संगम [1.5]; सूर्योदय-सूर्यास्त की सुंदरता, विवेकानंद चट्टान [1.5]; यात्रा की इच्छा/अनुभूति [1]।

Q21 [5]

Question: किसी एक विषय पर 120 शब्दों में अनुच्छेद लिखिए —
(क) खेलों का महत्त्व (ख) मोबाइल फ़ोन — वरदान या अभिशाप (ग) मेरी प्रिय पुस्तक

Expected answer / marking points: विषय-वस्तु [2], भाषा-शुद्धता [1.5], क्रमबद्धता/प्रस्तुति [1.5]।

Q22 [5]

Question: अपने विद्यालय के प्रधानाचार्य को पत्र लिखकर विद्यालय में खेल का पीरियड बढ़ाने का अनुरोध कीजिए।

ATHVA / OR

अपने दादाजी को पत्र लिखकर बताइए कि आपकी वार्षिक परीक्षा की तैयारी कैसी चल रही है।

Expected answer / marking points: औपचारिक पत्र-प्रारूप [1.5]; विषय-वस्तु — कारण, लाभ [2.5]; समापन, भाषा [1]।
अथवा: अनौपचारिक पत्र-प्रारूप [1.5]; विषय-वस्तु [2.5]; भाषा [1]।

Q23 **Question:** दो मित्रों के बीच “खेल और पढ़ाई में संतुलन” पर 80–100 शब्दों में संवाद लिखिए।
[5]

Expected answer / marking points: संवाद-प्रारूप [1]; विषय से जुड़े स्वाभाविक संवाद [3]; भाषा [1]।

Q24 **Question:** विद्यालय की “वार्षिक खेलकूद प्रतियोगिता” में भाग लेने के लिए छात्रों हेतु 40–50 शब्दों में सूचना लिखिए।
[5]

ATHVA / OR

अपने शिक्षक को ई-मेल लिखकर किसी कठिन अध्याय को दोबारा समझाने का अनुरोध कीजिए (60–80 शब्द)।

Expected answer / marking points: सूचना-प्रारूप [2]; विषय — तिथि, स्थान, पंजीकरण [2]; नाम/पद [1]।
अथवा: ई-मेल प्रारूप [2]; विषय-वस्तु [2]; समापन [1]।

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

English · Mock Paper 2 · Answer key

Friday, 26 February 2027 · 80 marks

Test ID: T2027-02-26

Class 9 · Reading · Grammar · Writing · Literature (Kaveri Units 1–8)

Q1 [10] **Question:** Read the passage and answer the questions.

> 1. Every year, millions of Indians donate blood, yet hospitals often report shortages. One unit of donated blood can help up to three patients, because it can be separated into red cells, plasma and platelets.
> 2. A healthy person between 18 and 65 years of age who weighs at least 45 kg can usually donate blood. The process takes about ten minutes, and the body replaces the lost fluid within a day. Red cells are replaced in a few weeks.
> 3. Blood cannot be manufactured in a factory. It must come from donors. Platelets last only about five days in storage, which is why regular donation is needed rather than large drives once a year.
> 4. Many myths stop people from donating — that it causes weakness, or that it is painful. In reality, donors feel only a small pinch, and they are given juice and a short rest afterwards.
> 5. Schools and colleges can help by spreading awareness. Students under 18 cannot donate, but they can encourage family members and help organise camps.

(i) How many patients can one unit of blood help? [1]
(ii) Into which three parts can donated blood be separated? [1]
(iii) What is the age range for donating blood? [1]
(iv) What is the minimum weight required? [1]
(v) How long does the donation process take? [1]
(vi) Why is regular donation needed? [1]
(vii) Name one myth about blood donation. [1]
(viii) What are donors given after donating? [1]
(ix) How can students under 18 help? [1]
(x) Find a word in paragraph 3 that means “made in large quantities by machines”. [1]

Expected answer / marking points: (i) Up to three. (ii) Red cells, plasma, platelets. (iii) 18 to 65 years. (iv) 45 kg. (v) About ten minutes. (vi) Blood cannot be manufactured and platelets last only about five days. (vii) It causes weakness / it is painful. (viii) Juice and a short rest. (ix) Spread awareness, encourage family members, help organise camps. (x) manufactured. [1 × 10]

Q2 **[10]**

Question: Read the passage and answer the questions.

> 1. Arjun had always been the fastest runner in his school. So when he lost the district 400 m race to a boy from a village school, he was shocked. That evening he refused to eat and threw his spikes into a corner.
> 2. His coach, an old athlete with a limp, came to see him the next morning. "Do you know why you lost?" he asked. Arjun shrugged. "You ran the first 200 metres for the crowd," the coach said. "The other boy ran the whole race for the finish line."
> 3. For the next three months, Arjun trained differently. He practised pacing himself, recorded his lap times in a notebook and went to bed early. He also began running with the village boy, Raju, on Sundays. Raju had no proper shoes, so Arjun gave him his old pair.
> 4. At the state meet, Arjun came second and Raju came first. On the podium, Arjun was surprised to find that he was smiling. He had lost the gold, but he had found a friend and learnt to run his own race.

(i) Why was Arjun shocked? [1]
(ii) How did he react that evening? [1]
(iii) According to the coach, why did Arjun lose? [2]
(iv) Write two changes Arjun made in his training. [2]
(v) How did Arjun help Raju? [1]
(vi) What was the result at the state meet? [1]
(vii) Find a word in paragraph 3 that means "controlling one's speed". [1]
(viii) What did Arjun learn by the end? [1]

Expected answer / marking points: (i) He lost the district 400 m race to a village-school boy though he was the fastest in his school [1]. (ii) Refused to eat and threw his spikes into a corner [1]. (iii) He ran the first 200 m to impress the crowd instead of running the whole race for the finish line [2]. (iv) Pacing himself, recording lap times, sleeping early, running with Raju (any two) [2]. (v) Gave him his old pair of shoes [1]. (vi) Raju first, Arjun second [1]. (vii) pacing [1]. (viii) To run his own race; friendship matters more than medals [1].

Q3 **[10]**

Question: Do as directed (1 mark each):
(i) Fill in with "much / many": How ___ books are there in the library?
(ii) Fill in with the correct form: If it rains, the match ___ (cancel).
(iii) Fill in with a modal: You ___ not worry; I will help you.
(iv) Change into passive: People celebrate Diwali all over India.
(v) Change into reported speech: The coach said, "You ran for the crowd."
(vi) Correct the error: She is more taller than her brother.
(vii) Combine using "so ... that": The tea was very hot. Grandpa could not drink it.
(viii) Fill in with the correct tense: While Ravi ___ (hide) behind the tree, the vendors surrounded him.
(ix) Change into reported speech: Ravi asked his mother, "Where is Grandpa?"
(x) Fill in with an article: Sentila made ___ unusual pot with a wide mouth.

Expected answer / marking points: (i) many (ii) will be cancelled (iii) need (iv) Diwali is celebrated all over India (by people). (v) The coach said that I/he had run for the crowd. (vi) She is taller than her brother. (vii) The tea was so hot that Grandpa could not drink it. (viii) was hiding (ix) Ravi asked his mother where Grandpa was. (x) an [1 × 10]

Q4 [5] **Question:** Write a descriptive paragraph (100–120 words) on "A Morning at a Railway Station".

ATHVA / OR

Write a speech (100–120 words) for the school assembly on "The Importance of Blood Donation".

Expected answer / marking points: Opening [1]; sights, sounds, people, activities with descriptive words [2.5]; closing; language [1.5].
OR: Greeting/opening [0.5]; points — need, eligibility, myths, how students can help [3]; conclusion + thanks; language [1.5].

Q5 [5] **Question:** Write a letter (100–120 words) to your friend telling him/her about a book that changed the way you think.

ATHVA / OR

Write a notice (50–60 words) for your school notice board about a blood donation awareness rally, as the Secretary of the Social Service Club.

Expected answer / marking points: Informal letter format [1.5]; content — book, what happened, how it changed your thinking [2.5]; language [1].
OR: Notice format — school name, NOTICE, date, heading, signature/designation [2]; details — what, when, where, who can join [2]; brevity/language [1].

Q6 [5] **Question:** Read the stanza and answer.

> The voice that wins its sunny way,
> A lonely home to cheer,
> Hath oft the fewest words to say;
> But, oh! those few, how dear!

(i) Whose voice "wins its sunny way"? [1]
(ii) What does such a voice do for a lonely home? [1]
(iii) What is special about the words of this voice? [1]
(iv) Pick out a pair of rhyming words. [1]
(v) What value does the stanza teach? [1]

Expected answer / marking points: (i) A kind, cheerful voice [1]. (ii) Brings cheer/happiness [1]. (iii) They are few but very dear/precious [1]. (iv) way–say / cheer–dear [1]. (v) Sincere, kind words matter more than many words [1].

Q7 **[5]**

Question: Read the stanza and answer.

> I cannot remember my mother,
> only sometimes in the midst of my play
> a tune seems to hover over my playthings,
> the tune of some song that she used to
> hum while rocking my cradle.

(i) When does the poet feel his mother's presence in this stanza? [1]
(ii) What does he remember — the song or the words? [1]
(iii) Which sense is used here? [1]
(iv) Is the setting indoor or outdoor? [1]
(v) What does "hover" suggest about the tune? [1]

Expected answer / marking points: (i) While playing [1]. (ii) Only the tune of a song [1]. (iii) Hearing (auditory) [1]. (iv) Indoor [1]. (v) The tune seems to float around him gently, like a presence [1].

Q8 **[2]**

Question: What made Avva cry when the narrator returned from the wedding, and what did she decide? (Unit 1)

Expected answer / marking points: She could not read the latest episode of Kashi Yatre herself and felt helpless [1]; she decided to learn the Kannada alphabet before Dassara [1].

Q9 **[2]**

Question: What did Mesoba tell the village council about Sentila? (Unit 2)

Expected answer / marking points: That Arenla had never refused to teach her; they wanted her to grow stronger after illness [1]; that soon Sentila would make the best pots in the village [1].

Q10 **[2]**

Question: Which pankhas are made in Bengal and from what? (Unit 3)

Expected answer / marking points: Delicate pankhas from the milky-white spongy centre of sola [1]; palm-leaf Tal Patar Pankha [1].

Q11 **[2]**

Question: What gift did Grandpa give Ravi and why was it funny? (Unit 4)

Expected answer / marking points: The Best Detective Stories [1]; Grandpa hinted that Ravi had been a poor "detective" who got fooled while trailing him [1].

Q12 **[2]**

Question: Why did Shruti decide to practise "one last time" in Act III? (Unit 6)

Expected answer / marking points: She loved performing the piece [1]; she planned to tell her group afterwards that she could not join the concert [1].

Q13 **[2]**

Question: Why did the mother say that "dreams remain dreams" for many people? (Unit 8)

Expected answer / marking points: People never move beyond wishful thinking or choose security [1]; circumstances like war or family needs change their lives [1].

Q14 [6] **Question:** Sudha Murty says, "For learning there is no age bar." How does Avva prove this? (120–150 words) (Unit 1)

ATHVA / OR

Compare Sentila and Shruti as young people following their passion. (120–150 words) (Units 2 and 6)

Expected answer / marking points: Avva at 62 decides to learn; works hard — reads, writes, recites [2]; reads Kashi Yatre on Dassara [1.5]; touches teacher's feet — humility; lesson about lifelong learning [1.5]; expression [1].
OR: Sentila — pot making against mother's wish, learns secretly, proves herself [2]; Shruti — fusion music, hides practice, finally speaks up [2]; both show determination; families finally accept [1]; expression [1].

Q15 [6] **Question:** How does "Vitamin-M" show that the elderly need respect, not pity? (120–150 words) (Unit 4)

ATHVA / OR

Describe the hardships of a Gramin Dak Sewak in the Thar desert and explain why Khetaram is grateful for his job. (120–150 words) (Unit 7)

Expected answer / marking points: Mother treats Grandpa like a child, loud tone [1.5]; Grandpa is independent, clever at chess, generous with his cap, remembers birthdays [2]; he outwits Ravi; lesson — respect their dignity and abilities [1.5]; expression [1].
OR: Heat near 50 °C, sandstorms, 20 km walks, 28 kg bag, no water to wash [2.5]; famine, one bajra crop, family of five; assured income even after 60 [2]; expression [1.5].

Q16 [6] **Question:** How do the pankha-makers and the Paralympians both show that tradition and courage can create new possibilities? (120–150 words) (Units 3 and 5)

Expected answer / marking points: Pankha-makers keep a craft alive through new designs, workshops and markets [2]; Paralympians break stereotypes, open new paths for others [2]; common idea — adapting with courage [1]; expression [1].

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

Maths · Mock Paper 3 · Answer key

Saturday, 27 February 2027 · 80 marks

Test ID: T2027-02-27

Class 9 · poora syllabus (Ch 1–14)

Q1 **[1]**

Question: The x-coordinate of every point on the y-axis is:

(a) 1 (b) 0

(c) –1 (d) equal to its y-coordinate

Expected answer / marking points: (b) 0

Q2 **[1]**

Question: Which point lies on the line y = x?

(a) (3, –3) (b) (3, 3)

(c) (0, 3) (d) (3, 0)

Expected answer / marking points: (b) (3, 3)

Q3 **[1]**

Question: A linear polynomial whose graph passes through the origin has the form:

(a) ax + b, b ≠ 0 (b) ax

(c) b (d) x^2

Expected answer / marking points: (b) ax

Q4 **[1]**

Question: If the line y = kx passes through (2, 8), then k =

(a) 2 (b) 4

(c) 8 (d) 16

Expected answer / marking points: (b) 4

Q5 **[1]**

Question: The sum of two irrational numbers is:

(a) always irrational (b) always rational

(c) may be rational or irrational (d) always an integer

Expected answer / marking points: (c) may be rational or irrational — $\sqrt{2} + (-\sqrt{2}) = 0$

Q6 **[1]**

Question: 0.454545... in the form p/q is:

(a) 45/100 (b) 5/11

(c) 9/20 (d) 4/9

Expected answer / marking points: (b) 5/11

Q7 **[1]**

Question: $x^2 - 1/9$ =

(a) $(x - 1/3)^2$ (b) $(x - 1/3)(x + 1/3)$

(c) $(x - 1/9)(x + 1/9)$ (d) $(x - 3)(x + 3)$

Expected answer / marking points: (b) $(x - 1/3)(x + 1/3)$

Q8 [1]

Question: $(x + y + z)^2$ =

(a) $x^2 + y^2 + z^2$

(b) $x^2 + y^2 + z^2 + 2xy + 2yz + 2zx$

(c) $x^2 + y^2 + z^2 + xy + yz + zx$

(d) $2x + 2y + 2z$

Expected answer / marking points: (b) $x^2 + y^2 + z^2 + 2xy + 2yz + 2zx$

Q9 [1]

Question: Of two unequal chords of a circle, the longer chord is:

(a) farther from the centre

(b) closer to the centre

(c) at the same distance

(d) outside the circle

Expected answer / marking points: (b) closer to the centre

Q10 [1]

Question: The area of a quarter circle of radius 14 cm is:

(a) 154 cm^2

(b) 616 cm^2

(c) 44 cm^2

(d) 308 cm^2

Expected answer / marking points: (a) 154 cm^2

Q11 [1]

Question: Three coins are tossed together. The number of outcomes in the sample space is:

(a) 3

(b) 6

(c) 8

(d) 9

Expected answer / marking points: (c) 8

Q12 [1]

Question: The 7th triangular number is:

(a) 21

(b) 28

(c) 36

(d) 49

Expected answer / marking points: (b) 28

Q13 [1]

Question: The 8th term of the GP 3, –3, 3, ... is:

(a) 3

(b) –3

(c) 24

(d) –24

Expected answer / marking points: (b) –3

Q14 [1]

Question: A counterexample to “Every multiple of 3 is odd” is:

(a) 9

(b) 15

(c) 6

(d) 21

Expected answer / marking points: (c) 6

Q15 [1]

Question: A 100% stacked bar chart compares:

(a) absolute values only

(b) proportions

(c) only totals

(d) time only

Expected answer / marking points: (b) proportions

Q16 [1] **Question:** gcd(81, 27) =

(a) 3 (b) 9

(c) 27 (d) 81

Expected answer / marking points: (c) 27

Q17 [1] **Question:** A median of a triangle joins a vertex to:

(a) the opposite vertex (b) the midpoint of the opposite side

(c) the centroid only (d) the circumcentre

Expected answer / marking points: (b) the midpoint of the opposite side

Q18 [1] **Question:** The volume of a cone of radius 3 cm and height 7 cm is ($\pi = 22/7$):

(a) 66 cm^3 (b) 198 cm^3

(c) 22 cm^3 (d) 132 cm^3

Expected answer / marking points: (a) 66 cm^3

Q19 [1] **Question: A:** The line $3x - 4y = 12$ cuts the x-axis at (4, 0).
R: At every point on the x-axis, $y = 0$.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q20 [1] **Question: A:** The probability of an event can be 1.2.
R: For any event E, $0 \le P(E) \le 1$.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (d)

Q21 [2] **Question:** Show that A(–1, –1), B(–1, 4) and C(3, –1) form a right-angled triangle and find its area.

Expected answer / marking points: AB vertical (length 5), AC horizontal (length 4) ⇒ right angle at A [1]; area = ½ × 5 × 4 = 10 sq units [1].

Q22 [2] **Question:** Factorise $a^2 + 2ab + b^2 - 16$.

Expected answer / marking points: $(a + b)^2 - 4^2$ [1] = $(a + b - 4)(a + b + 4)$ [1].

Q23 [2] **Question:** A wheel has diameter 56 cm. How many complete revolutions does it make to cover 1.76 km?

Expected answer / marking points: Circumference = 22/7 × 56 = 176 cm [1]; 1,76,000 ÷ 176 = 1000 revolutions [1].

Q24 [2] **Question:** A month of the year is chosen at random. Find the probability that (i) it has 31 days (ii) its name starts with J.

Expected answer / marking points: (i) 7/12 [1] (ii) January, June, July ⇒ 3/12 = 1/4 [1].

Q25 [2] **Question:** How many terms are there in the AP 7, 13, 19, ..., 205?

Expected answer / marking points: 7 + 6(n – 1) = 205 [1] ⇒ n – 1 = 33 ⇒ n = 34 [1].

Q26 [3] **Question:** Express 1.272727... in the form p/q.

Expected answer / marking points: x = 1.2727...; 100x = 127.2727... [1]; 99x = 126 [1] ⇒ x = 126/99 = 14/11 [1].

Q27 [3] **Question:** A triangular park has sides 120 m, 80 m and 50 m. Find its area. Fencing is done leaving 3 m for a gate, at ₹20 per m. Find the cost.

Expected answer / marking points: s = 125; area = $\sqrt{(125 \times 5 \times 45 \times 75)} \approx 1452.37$ m^2 [1.5]; fencing = 250 – 3 = 247 m; cost ₹4940 [1.5].

Q28 [3] **Question:** BC is a chord of a circle with centre O, and A is a point on the major arc with ∠BAC = 30°. Find ∠BOC and show that BC equals the radius.

Expected answer / marking points: ∠BOC = 2 × 30° = 60° [1]; OB = OC ⇒ ∠OBC = ∠OCB = 60° [1]; △BOC is equilateral ⇒ BC = OB = radius [1].

Q29 [3] **Question:** A bag has 1 red and 2 green balls. Two balls are drawn one after another without replacement. Using a tree diagram, find P(both green), P(red first) and P(one red and one green).

Expected answer / marking points: P(GG) = 2/3 × 1/2 = 1/3 [1]; P(R first) = 1/3 [1]; P(RG or GR) = 1/3 × 1 + 2/3 × 1/2 = 2/3 [1].

Q30 [3] **Question:** Write the steps of an algorithm to find the LCM of two numbers using their gcd. Use it to find LCM(18, 24).

Expected answer / marking points: Find gcd by Euclid's method; LCM = (a × b) ÷ gcd [1.5]; gcd(18, 24) = 6 ⇒ LCM = 432 ÷ 6 = 72 [1.5].

Q31 [3] **Question:** Show that the line segments joining the midpoints of the opposite sides of a quadrilateral bisect each other.

Expected answer / marking points: Midpoints P, Q, R, S of AB, BC, CD, DA form a parallelogram PQRS (Midpoint Theorem applied with diagonals) [2]; PR and QS are its diagonals, and diagonals of a parallelogram bisect each other [1].

Q32 [5] **Question:** Draw the graphs of y = 2x – 4 and y = –x + 5. Find their point of intersection and the area of the triangle formed by the two lines and the y-axis.

ATHVA / OR

A taxi charges a fixed amount plus a rate per km. The fare for 10 km is ₹105 and for 15 km is ₹155. Find the fixed charge and the rate per km. What will 25 km cost?

Expected answer / marking points: Graph [2]; 2x – 4 = –x + 5 ⇒ (3, 2) [1]; y-intercepts –4 and 5 ⇒ base 9 [1]; area = ½ × 9 × 3 = 13.5 sq units [1].
OR: f + 10c = 105, f + 15c = 155 [1.5]; 5c = 50 ⇒ c = ₹10, f = ₹5 [2]; 25 km: 5 + 250 = ₹255 [1.5].

Q33 [5] **Question:** (i) If $x + 1/x = 4$, find $x^2 + 1/x^2$ and $x^3 + 1/x^3$. (ii) Simplify $[(a^2 - b^2)^3 + (b^2 - c^2)^3 + (c^2 - a^2)^3] \div [(a - b)^3 + (b - c)^3 + (c - a)^3]$.

ATHVA / OR

(i) Expand $(2a - 3b + c)^2$. (ii) Evaluate 1002^2 and 998^2 using identities, and hence find $1002^2 - 998^2$.

Expected answer / marking points: (i) $16 - 2 = 14$ [1]; $(x + 1/x)^3 - 3(x + 1/x) = 64 - 12 = 52$ [1.5]. (ii) In both, the three terms add to 0, so each sum = 3 × product [1]; $= 3(a^2 - b^2)(b^2 - c^2)(c^2 - a^2) \div 3(a - b)(b - c)(c - a) = (a + b)(b + c)(c + a)$ [1.5].
OR: (i) $4a^2 + 9b^2 + c^2 - 12ab - 6bc + 4ca$ [2]. (ii) 1004004 and 996004 [2]; difference 8000 (or (1002 + 998)(4)) [1].

Q34 [5] **Question:** Prove that the opposite angles of a cyclic quadrilateral add up to 180°.

Expected answer / marking points: Figure: ABCD cyclic, centre O; join OB, OD [1]. Arc BCD subtends $\angle BOD = 2\angle A$ at the centre, and arc BAD subtends reflex $\angle BOD = 2\angle C$ [2]. $2\angle A + 2\angle C = 360^\circ$ [1] $\Rightarrow \angle A + \angle C = 180^\circ$; similarly $\angle B + \angle D = 180^\circ$ [1].

Q35 [5] **Question:** A cuboidal water tank is 6 m long, 5 m wide and 4.5 m deep. (i) How many litres does it hold? (ii) A building of 30 people uses 150 L per person per day. For how many days will a full tank last? (iii) Find the cost of painting its four inner walls at ₹10 per m^2.

Expected answer / marking points: (i) $6 \times 5 \times 4.5 = 135\ m^3 = 1{,}35{,}000$ L [1.5]. (ii) $30 \times 150 = 4500$ L/day $\Rightarrow$ 30 days [1.5]. (iii) $2(6 + 5) \times 4.5 = 99\ m^2 \Rightarrow$ ₹990 [2].

Q36 [4] **Question:**

> A new tiffin service on a food app received 50 ratings with an average of 4.2 stars. The next 25 ratings had an average of 3.6 stars.

(i) Find the total stars in the first 50 ratings. [1]
(ii) Find the overall average after 75 ratings. [1]
(iii) How many more 5-star ratings are needed to bring the average back to 4.2 or more? [2]

Expected answer / marking points: (i) 210 [1]. (ii) $(210 + 90)/75 = 4.0$ [1]. (iii) $(300 + 5k)/(75 + k) \geq 4.2 \Rightarrow 0.8k \geq 15 \Rightarrow k \geq 18.75$ [1] $\Rightarrow$ 19 ratings [1].

Q37 [4] **Question:**

> Meera saves ₹10 on day 1, ₹20 on day 2, ₹30 on day 3, and so on. Her friend saves ₹1 on day 1, ₹2 on day 2, ₹4 on day 3, doubling every day.

(i) How much does Meera save on day 15? [1]
(ii) Using triangular numbers, find Meera's total savings in the first 15 days. [1]
(iii) On which day does the friend's daily saving first become more than Meera's daily saving? Show working. [2]

Expected answer / marking points: (i) ₹150 [1]. (ii) $10 \times (15 \times 16/2) =$ ₹1200 [1]. (iii) Meera 10n, friend 2^{n-1}: day 7 — 70 vs 64; day 8 — 80 vs 128 [1] $\Rightarrow$ day 8 [1].

Q38 **Question:**

[4]

> An ice-cream cone is a cone of radius 3 cm and height 10 cm, completely filled, with a hemisphere of ice-cream of the same radius on top. (π = 22/7)

(i) Find the volume of ice-cream in the cone part. [1]
(ii) Find the volume of the hemispherical top. [1]
(iii) How many such full cones can be made from a 3-litre tub of ice-cream? [2]

Expected answer / marking points: (i) ⅓ × 22/7 × 9 × 10 ≈ 94.29 cm^3 [1]. (ii) ⅔ × 22/7 × 27 ≈ 56.57 cm^3 [1]. (iii) Total ≈ 150.86 cm^3; 3000 ÷ 150.86 ≈ 19.9 ⇒ 19 full cones [2].

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

Maths · Full Paper 2 (Final rehearsal) · Answer key

Sunday, 28 February 2027 · 80 marks

Test ID: T2027-02-28

Class 9 · Ganita Manjari Part I + II — poora syllabus (Ch 1–14)

Q1 **[1]**

Question: The distance of the point (–6, 8) from the x-axis is:

(a) 6 (b) 8

(c) 10 (d) 14

Expected answer / marking points: (b) 8

Q2 **[1]**

Question: The line joining (2, 3) and (2, –5) is parallel to the:

(a) x-axis (b) y-axis

(c) line y = x (d) none of these

Expected answer / marking points: (b) y-axis

Q3 **[1]**

Question: Which of the following is NOT a polynomial?

(a) $x^2 - 2x$ (b) $\sqrt{x} + 1$

(c) 3 (d) $y^3 - y$

Expected answer / marking points: (b) $\sqrt{x} + 1$

Q4 **[1]**

Question: The y-intercept of the line y = 2x + 3 is:

(a) 2 (b) 3

(c) –3 (d) 1.5

Expected answer / marking points: (b) 3

Q5 **[1]**

Question: A rational number between $\sqrt{2}$ and $\sqrt{3}$ is:

(a) 1.2 (b) 1.5

(c) 1.8 (d) 2

Expected answer / marking points: (b) 1.5 — 1.414 < 1.5 < 1.732

Q6 **[1]**

Question: $(a - b)^2 + (a + b)^2$ =

(a) $2a^2 + 2b^2$ (b) 4ab

(c) $2a^2 - 2b^2$ (d) $a^2 + b^2$

Expected answer / marking points: (a) $2a^2 + 2b^2$

Q7 **[1]**

Question: If x + y = 10 and xy = 21, then $x^2 + y^2$ =

(a) 79 (b) 58

(c) 100 (d) 42

Expected answer / marking points: (b) 58

Q8 **Question:** Two chords of a circle that are equidistant from the centre are:
[1]

(a) perpendicular (b) parallel
(c) equal (d) unequal

Expected answer / marking points: (c) equal

Q9 **Question:** ABCD is a cyclic quadrilateral with $\angle A = 75^\circ$. Then $\angle C$ =
[1]

(a) 75° (b) 105°
(c) 15° (d) 115°

Expected answer / marking points: (b) 105°

Q10 **Question:** The area of a triangle with base 10 cm and height 6 cm is:
[1]

(a) 60 cm^2 (b) 30 cm^2
(c) 16 cm^2 (d) 32 cm^2

Expected answer / marking points: (b) 30 cm^2

Q11 **Question:** A card is drawn from cards numbered 1 to 20. The probability that it is a multiple of 5 is:
[1]

(a) 1/4 (b) 1/5
(c) 1/10 (d) 1/20

Expected answer / marking points: (b) 1/5

Q12 **Question:** The common difference of the AP 10, 7, 4, ... is:
[1]

(a) 3 (b) –3
(c) 7 (d) –7

Expected answer / marking points: (b) –3

Q13 **Question:** The 6th term of the GP 1, 2, 4, ... is:
[1]

(a) 12 (b) 16
(c) 32 (d) 64

Expected answer / marking points: (c) 32

Q14 **Question:** Which of the following is a proposition?
[1]

(a) Close the window. (b) What a lovely day!
(c) 7 is a prime number. (d) Is it raining?

Expected answer / marking points: (c) 7 is a prime number.

Q15 **Question:** The mean of the first five even natural numbers is:
[1]

(a) 5 (b) 6
(c) 8 (d) 10

Expected answer / marking points: (b) 6

Q16 [1] **Question:** In a 100% stacked bar chart, all the bars:

(a) have different heights (b) have the same total height

(c) start from different levels (d) show only one category

Expected answer / marking points: (b) have the same total height

Q17 [1] **Question:** gcd(96, 36) =

(a) 6 (b) 12

(c) 18 (d) 24

Expected answer / marking points: (b) 12

Q18 [1] **Question:** The volume of a cylinder with radius 7 cm and height 2 cm is:

(a) 154 cm^3 (b) 308 cm^3

(c) 88 cm^3 (d) 616 cm^3

Expected answer / marking points: (b) 308 cm^3

Q19 [1] **Question: A:** The figure formed by joining the midpoints of the sides of a rectangle (in order) is a parallelogram.
R: The figure formed by joining the midpoints of the sides of any quadrilateral (in order) is a parallelogram.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q20 [1] **Question: A:** The point (2, 3) lies on the graph of $3x - 2y = 0$.
R: Every solution of a linear equation in two variables is a point on its graph.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q21 [2] **Question:** Find the perimeter of the triangle with vertices (0, 0), (4, 0) and (0, 3).

Expected answer / marking points: Sides 4 and 3 along axes, hypotenuse $\sqrt{(16 + 9)} = 5$ [1]; perimeter = 12 units [1].

Q22 [2] **Question:** If $x - 1/x = 3$, find $x^2 + 1/x^2$.

Expected answer / marking points: $(x - 1/x)^2 = x^2 + 1/x^2 - 2$ [1] $\Rightarrow 9 + 2 = 11$ [1].

Q23 [2] **Question:** Find the area of a sector of radius 10.5 cm and central angle 60°.

Expected answer / marking points: $60/360 \times 22/7 \times 10.5^2$ [1] = 57.75 cm^2 [1].

Q24 [2] **Question:** A letter is chosen at random from the word PROBABILITY. Find the probability that it is (i) B (ii) a vowel.

Expected answer / marking points: 11 letters [0.5]; (i) 2/11 [0.5]; (ii) O, A, I, I ⇒ 4/11 [1].

Q25 [2] **Question:** Write the explicit and the recursive formula for 2, 10, 50, 250, ...

Expected answer / marking points: Explicit: $t_n = 2 \times 5^{n-1}$ [1]; recursive: $t_1 = 2$, $t_n = 5t_{n-1}$ [1].

Q26 [3] **Question:** Given that $\sqrt{3}$ is irrational, prove that $2 + \sqrt{3}$ is irrational.

Expected answer / marking points: Suppose $2 + \sqrt{3} = r$, rational [1]; then $\sqrt{3} = r - 2$, which is rational (difference of rationals) [1]; contradiction, so $2 + \sqrt{3}$ is irrational [1].

Q27 [3] **Question:** The sides of a triangle are in the ratio 3 : 4 : 5 and its perimeter is 144 cm. Find its area using Heron's formula.

Expected answer / marking points: Sides 36, 48, 60 cm [1]; s = 72 [0.5]; $\sqrt{(72 \times 36 \times 24 \times 12)} = \sqrt{746496} = 864\ \text{cm}^2$ [1.5].

Q28 [3] **Question:** In an AP, the 3rd term is 16 and the 7th term exceeds the 5th term by 12. Find the AP.

Expected answer / marking points: $t_7 - t_5 = 2d = 12 \Rightarrow d = 6$ [1]; $a + 2d = 16 \Rightarrow a = 4$ [1]; AP: 4, 10, 16, 22, ... [1].

Q29 [3] **Question:** Solve: $x/2 + y/3 = 4$ and $2x - 3y = -10$.

Expected answer / marking points: First ⇒ $3x + 2y = 24$ [0.5]; ×3 and ×2: $9x + 6y = 72$, $4x - 6y = -20$ [1]; $13x = 52 \Rightarrow x = 4$ [1]; $y = 6$ [0.5].

Q30 [3] **Question:** Prove that the perpendicular drawn from the centre of a circle to a chord bisects the chord.

Expected answer / marking points: OM ⊥ AB; join OA, OB [0.5]; in right △OMA and △OMB: OA = OB (radii), OM common [1]; RHS ⇒ congruent [1] ⇒ AM = BM [0.5].

Q31 [3] **Question:** 3 L of a 40% juice drink is mixed with 2 L of a 15% juice drink. Find the juice concentration of the mixture. How much water must be added to make it 25%?

Expected answer / marking points: Juice = 1.2 + 0.3 = 1.5 L in 5 L ⇒ 30% [1.5]; $1.5/(5 + w) = 0.25 \Rightarrow w = 1$ L [1.5].

Q32 [5] **Question:** Draw the graphs of $y = x$ and $x + y = 4$ on the same axes. Find (i) their point of intersection (ii) the vertices and area of the triangle formed by these lines and the x-axis.

ATHVA / OR

The sum of the digits of a two-digit number is 9. If 27 is subtracted from the number, the digits are reversed. Find the number.

Expected answer / marking points: Tables + graph [2]; (i) (2, 2) [1]; (ii) vertices (0, 0), (4, 0), (2, 2) [1]; area = ½ × 4 × 2 = 4 sq units [1].
OR: Let tens digit x, units y: $x + y = 9$ [1]; $(10x + y) - 27 = 10y + x \Rightarrow x - y = 3$ [2]; $x = 6$, $y = 3 \Rightarrow 63$ [1.5]; check $63 - 27 = 36$ ✓ [0.5].

Q33 [5] **Question:** In △ABC, AD is a median and E is the midpoint of AD. BE is produced to meet AC at F. Prove that AF = ⅓ AC.

ATHVA / OR

Prove that a diagonal of a parallelogram divides it into two congruent triangles. Hence show that its opposite angles are equal.

Expected answer / marking points: Draw DG ∥ BF meeting AC at G [1]. In △ADG, E is the midpoint of AD and EF ∥ DG ⇒ F is the midpoint of AG, so AF = FG (converse of Midpoint Theorem) [1.5]. In △BCF, D is the midpoint of BC and DG ∥ BF ⇒ G is the midpoint of FC, so FG = GC [1.5]. Hence AF = FG = GC = ⅓AC [1].
OR: Diagonal AC: AB ∥ DC ⇒ ∠BAC = ∠DCA; AD ∥ BC ⇒ ∠DAC = ∠BCA; AC common [2]; ASA ⇒ △ABC ≅ △CDA [1]; CPCT ⇒ ∠B = ∠D [1]; similarly with diagonal BD ⇒ ∠A = ∠C [1].

Q34 [5] **Question:** A cone of height 24 cm and base radius 6 cm, made of clay, is reshaped into a sphere. Find (i) the radius of the sphere (ii) its surface area (iii) the total surface area of one hemisphere if the sphere is cut into two equal halves. (π = 22/7)

Expected answer / marking points: (i) $\frac{1}{3}\pi \times 36 \times 24 = \frac{4}{3}\pi r^3 \Rightarrow r^3 = 216 \Rightarrow r = 6$ cm [2]. (ii) $4 \times \frac{22}{7} \times 36 \approx 452.57$ cm^2 [1.5]. (iii) $3\pi r^2 = 3 \times \frac{22}{7} \times 36 \approx 339.43$ cm^2 [1.5].

Q35 [5] **Question:** Rani deposits ₹1000 in a scheme where her money doubles every 6 years. Rahul deposits ₹1000 in a box and adds ₹1000 more every 6 years. (i) Write the amounts for both after 6, 12 and 18 years. (ii) Which is a GP and which is an AP? (iii) After how many years will Rani's amount first exceed ₹50,000?

Expected answer / marking points: (i) Rani: 2000, 4000, 8000; Rahul: 2000, 3000, 4000 [2]. (ii) Rani — GP (r = 2); Rahul — AP (d = 1000) [1]. (iii) $1000 \times 2^n > 50000 \Rightarrow 2^n > 50 \Rightarrow n = 6$ periods (64000) ⇒ 36 years [2].

Q36 [4] **Question:**

> In a treasure hunt, a team starts at S(0, 0), walks 5 units east to A, then 3 units north to B, then 8 units west to C.

(i) Write the coordinates of C. [1]
(ii) In which quadrant is C? [1]
(iii) Find the total length of the path walked and the straight-line distance SB. [2]

Expected answer / marking points: (i) A(5, 0), B(5, 3), C(−3, 3) [1]. (ii) II quadrant [1]. (iii) 5 + 3 + 8 = 16 units [1]; SB = $\sqrt{(25 + 9)} = \sqrt{34} \approx 5.83$ units [1].

Q37 [4] **Question:**

> A school notice says: "If it rains, the assembly will be held in the hall."

(i) Write the converse of this proposition. [1]
(ii) One day the assembly was held in the hall. Can we be sure it rained? Why? [1]
(iii) Give a counterexample to "If a number is divisible by 4, then it is divisible by 8." Is its converse true? [2]

Expected answer / marking points: (i) If the assembly is held in the hall, then it rains/rained [1]. (ii) No — the converse need not be true; the hall may be used for other reasons (heat, exams) [1]. (iii) 12 is divisible by 4 but not by 8 [1]; converse "If divisible by 8, then divisible by 4" is true (8 = 4 × 2) [1].

Q38 **Question:**

[4]

> A spinner has 8 equal sectors numbered 1 to 8.

(i) Find P(even number). [1]
(ii) Find P(prime number). [1]
(iii) Find P(number greater than 6 or equal to 1). If the spinner is spun 80 times, about how many times would you expect this event? [2]

Expected answer / marking points: (i) 4/8 = 1/2 [1]. (ii) 2, 3, 5, 7 ⇒ 4/8 = 1/2 [1]. (iii) {1, 7, 8} ⇒ 3/8 [1]; 80 × 3/8 = 30 times [1].

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

Parent Sunday handover checklist

- [ ] Agle Monday–Sunday ka student pack print karke diya.
- [ ] Parent solved key alag rakhi; student copy ke saath share nahi ki.
- [ ] Pichhle week ka pending kaam aur 3 weak topics review kiye.
- [ ] School exam date sheet aur tuition timings se schedule match kiya.

Agli priority / teacher se poochhna: